Capture the Flag 6: Active or Inactive?

Goal: Make your Student Directory respond to student data by conditionally showing, hiding, and changing UI elements based on each student's state.

Instructor Note

These activities require your own problem-solving and research skills.
DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may **NOT** ask AI to generate the solution/code for a flag.

🏴 Flag 0 — Create Your Feature Branch

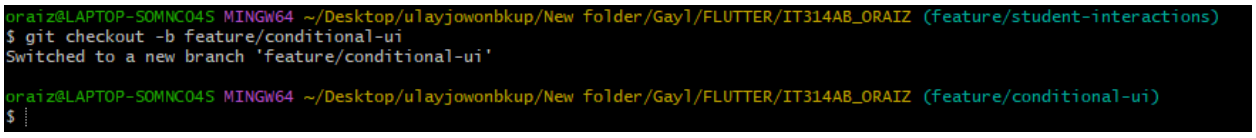
Objective: Create a new feature branch before adding conditional UI.

Create:

`feature/conditional-ui`

Screenshot:

- Terminal showing your current branch



🏴 Flag 1 — Show Active or Inactive

Objective: Change the Student Card based on whether a student is active or inactive.

Add an `active` status to your student data.

Research how Flutter uses **conditional expressions** to display different widgets.

Example states:

- Active
- Inactive

Test

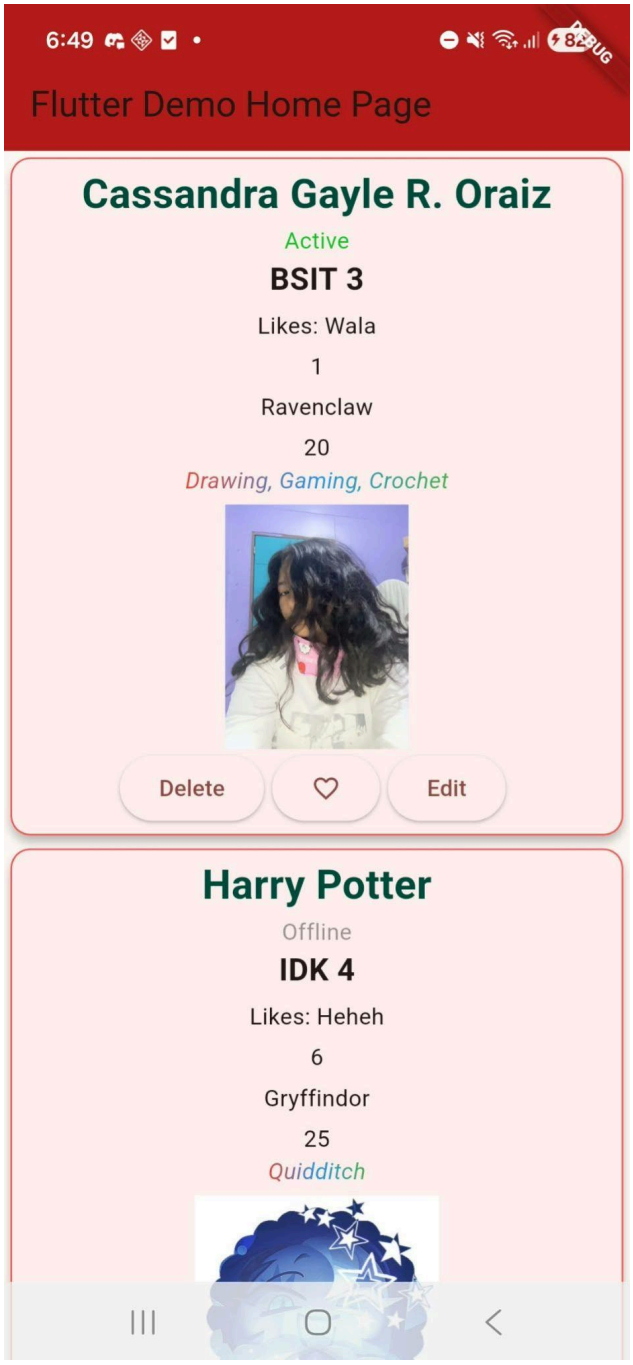
Active students should display an Active indicator.

Inactive students should display an Inactive indicator.

Screenshot:

- Active Student Card
- Inactive Student Card
- Conditional logic in your code

```
student.isActive
    ? Text(
        'Offline',
        style: TextStyle(color: Color.fromARGB(255, 158, 148, 148)),
      ) // Text
    : Text(
        'Active',
        style: TextStyle(color: Color.fromARGB(255, 14, 199, 39)),
      ), // Text
```


6:49
DEBUG
Flutter Demo Home Page
Cassandra Gayle R. Oraiz
Active
BSIT 3
Likes: Wala
1
Ravenclaw
20
Drawing, Gaming, Crochet
Delete
Edit
Harry Potter
Offline
IDK 4
Likes: Heheh
6
Gryffindor
25
Quidditch

🏴 Flag 2 — Show or Hide

Objective: Display certain UI elements only when a condition is true.

Add a conditional element to the Student Card.

Examples:

- Favorite badge appears only when favorited
- Warning appears only when inactive
- Delete action appears only if active

Research ways to conditionally include or exclude widgets from the widget tree.

Test

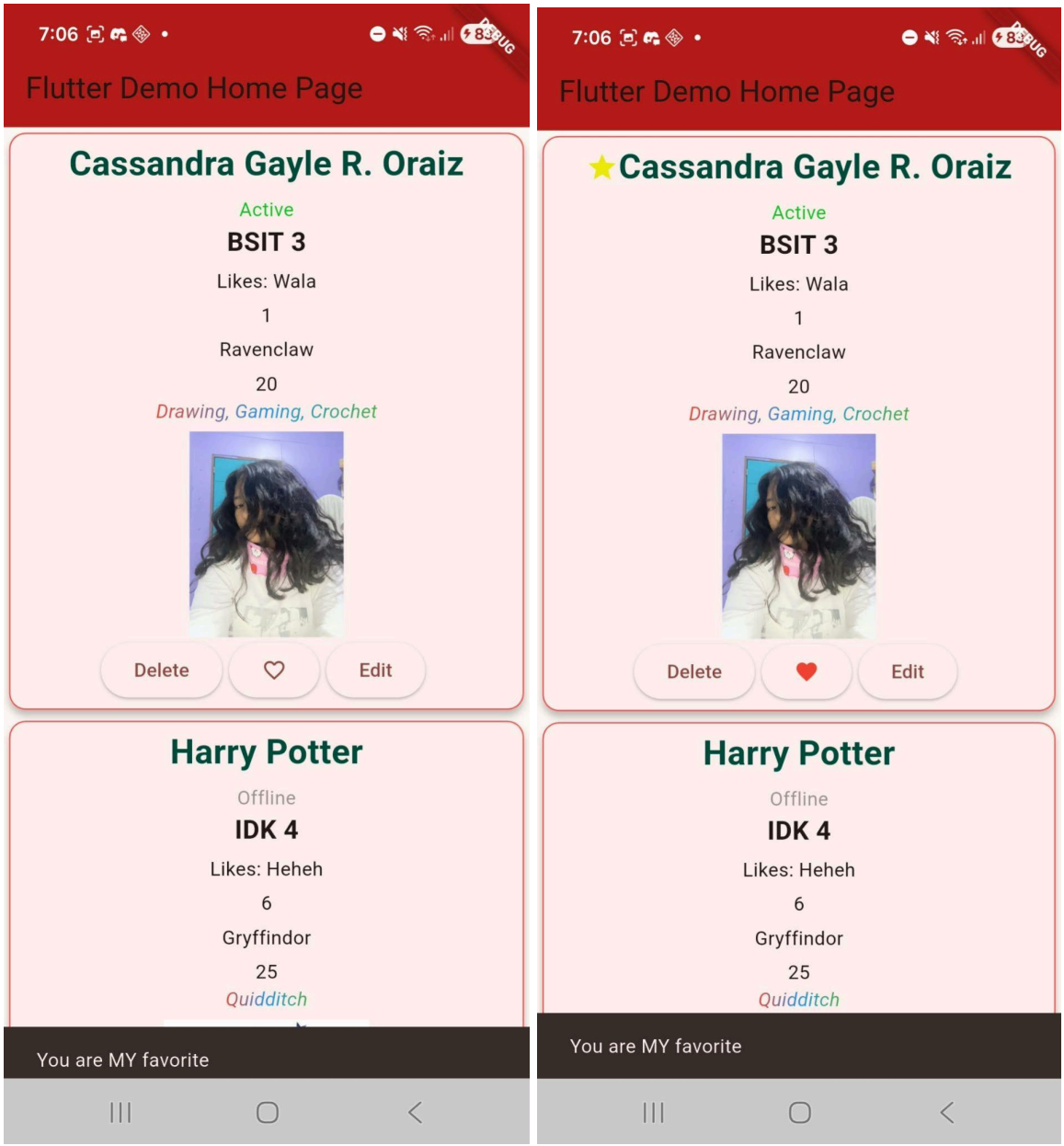
When the condition is true, the element appears.

When the condition is false, the element disappears.

Screenshot:

- UI with the element visible
- UI with the element hidden
- Conditional widget code

```
children: [
  SizedBox(height: 5),
  Row(
    mainAxisAlignment: .center,
    children: [
      student.fav
          ? Icon(
              Icons.star,
              color: const Color.fromARGB(255, 238, 234, 12),
            ) // Icon
          : SizedBox.shrink(),
      Text(
        student.name,
        style: TextStyle(
          fontSize: 25.0,
          color: Colors.teal.shade900,
          fontWeight: FontWeight.bold,
        ), // TextStyle
      ), // Text
    ],
  ), // Row
```



The star is linked to the heart/favorite button. If heart is red star appears else it's not there.

🏴 Flag 3 — The Empty Directory

Objective: Handle situations where there are no students to display.

Modify your Student Directory so it detects when the student list is empty.

Create an **empty state**.

Examples:

- "No students found"
- Empty-state icon
- Button to add a student

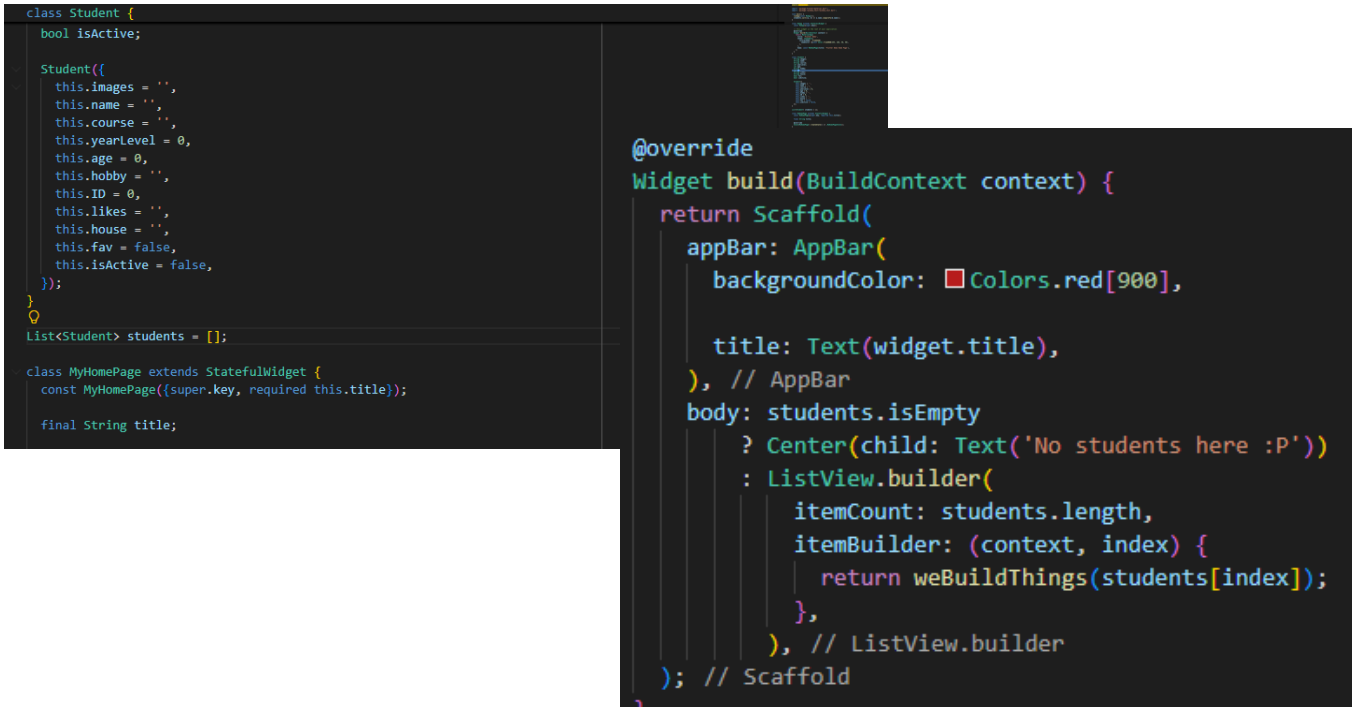
Test

Delete all students in your application (not the code)

Instead of showing a blank screen, your application should display your empty state.

Screenshot:

- Empty student list
- Empty-state UI
- Condition checking for an empty list



```
class Student {
  bool isActive;

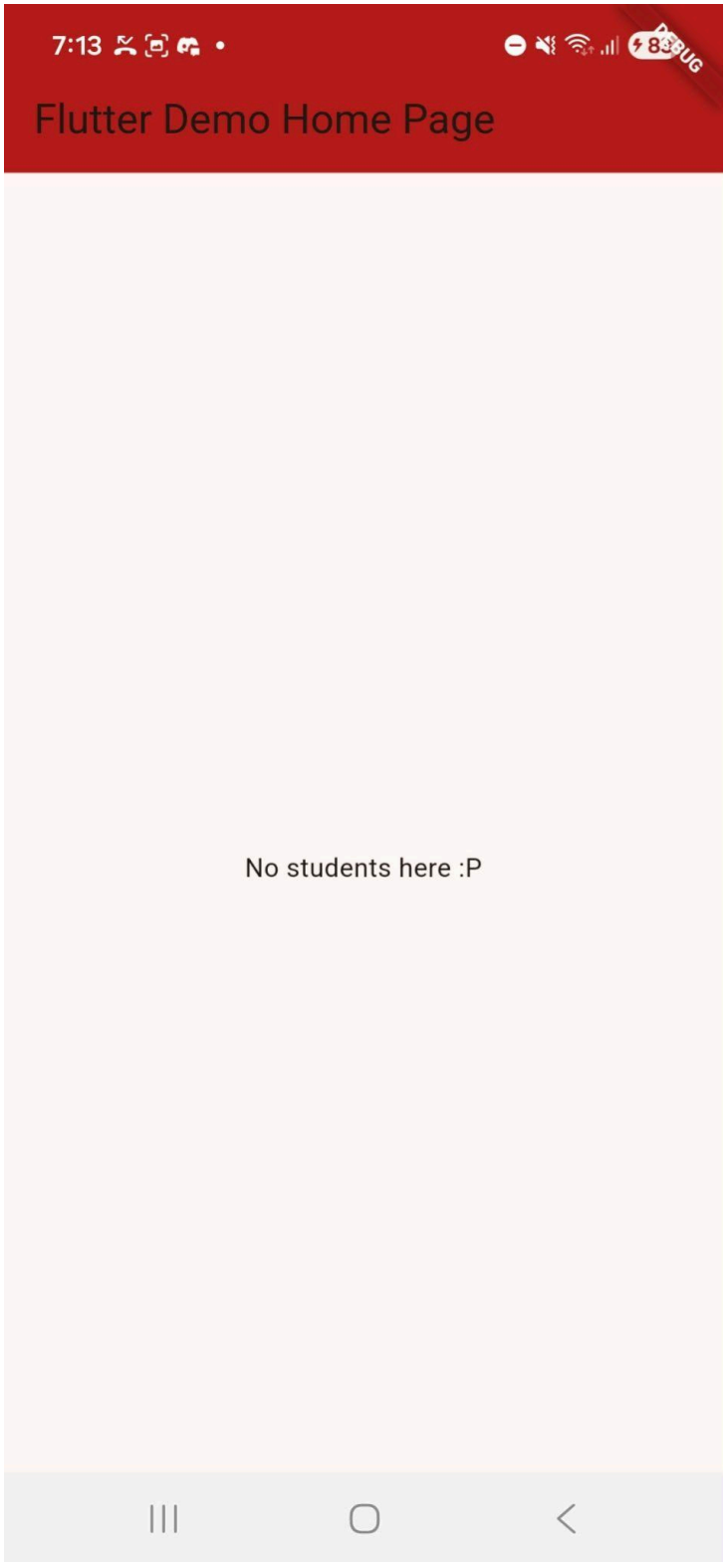
  Student({
    this.images = '',
    this.name = '',
    this.course = '',
    this.yearLevel = 0,
    this.age = 0,
    this.hobby = '',
    this.ID = 0,
    this.likes = '',
    this.house = '',
    this.fav = false,
    this.isActive = false,
  });
}
List<Student> students = [];

class MyHomePage extends StatefulWidget {
  const MyHomePage({super.key, required this.title});

  final String title;
```

```
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      backgroundColor: Colors.red[900],

      title: Text(widget.title),
    ), // AppBar
    body: students.isEmpty
        ? Center(child: Text('No students here :P'))
        : ListView.builder(
            itemCount: students.length,
            itemBuilder: (context, index) {
              return weBuildThings(students[index]);
            },
          ), // ListView.builder
  ); // Scaffold
```


7:13
DEBUG
Flutter Demo Home Page
No students here :P

🏴 Flag 4 — Loading the Directory

Objective: Display different UI while student data is being loaded.

Create a loading state for your Student Directory.

Research how Flutter can conditionally display a loading indicator.

Examples:

- `CircularProgressIndicator`
- Loading message
- Loading placeholder

Test

When the application is in a loading state, the student list should not be displayed as the main content.

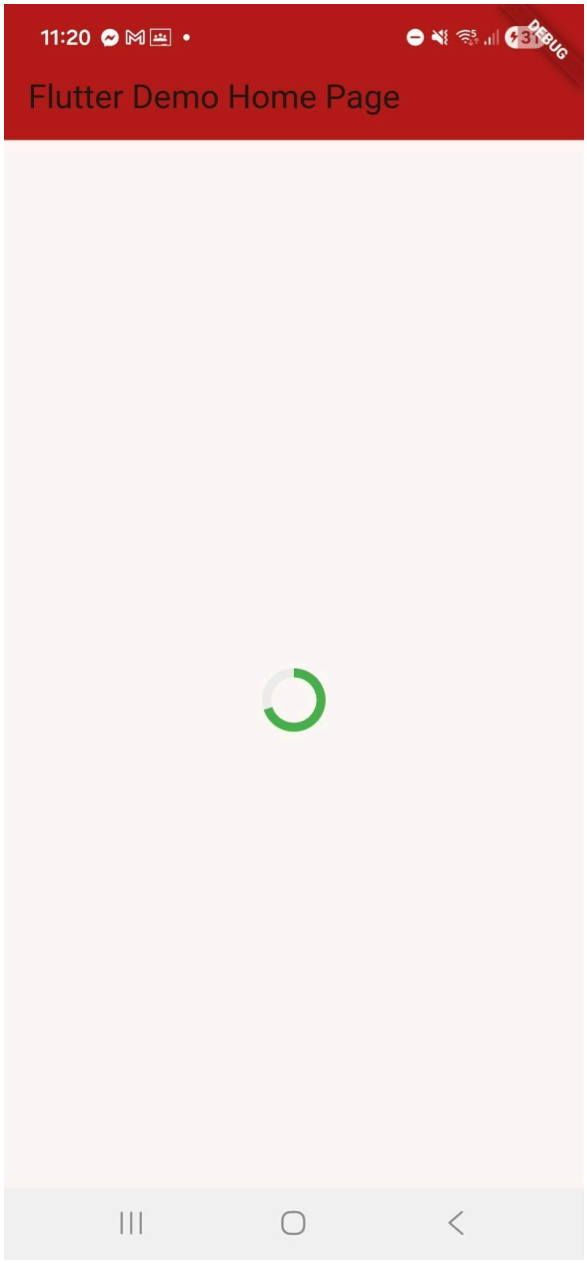
Once loading is complete, the Student Directory should appear.

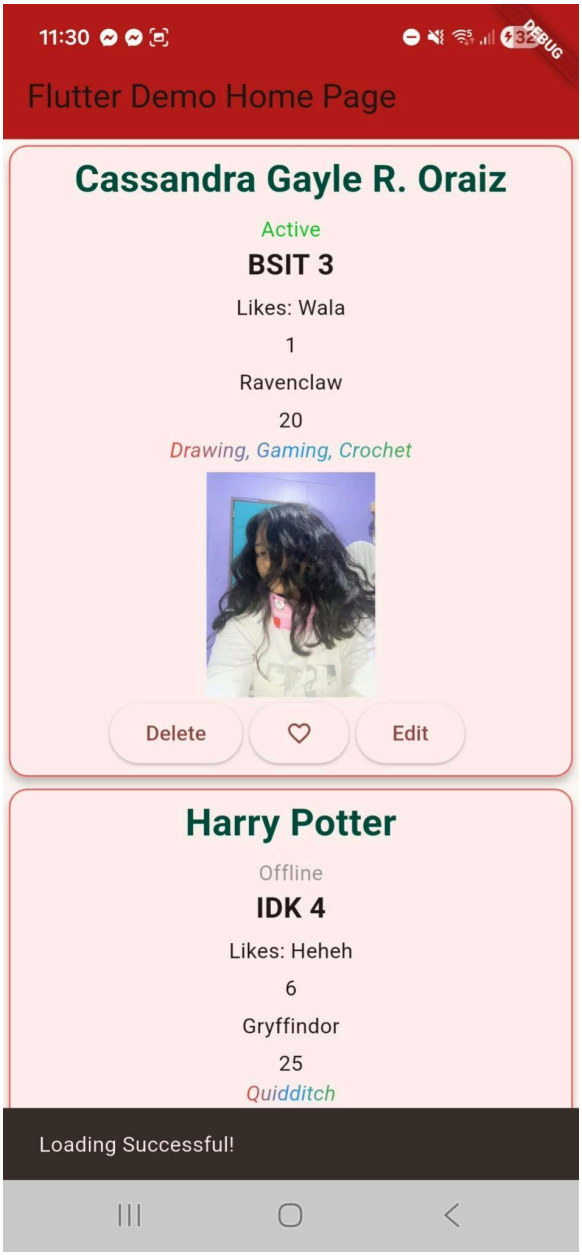
Screenshot:

- Loading state
- Loaded state
- Conditional loading logic

```
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      backgroundColor: Colors.red[900],

      title: Text(widget.title),
    ), // AppBar
    body: isLoading
        ? Center(
            child: CircularProgressIndicator(
              value: 0.7,
              color: Colors.green,
              backgroundColor: Colors.grey[200],
              strokeWidth: 6.0,
            ), // CircularProgressIndicator
          ) // Center
        : students.isEmpty
        ? Center(child: Text('No students here :P'))
        : ListView.builder(
            itemCount: students.length,
            itemBuilder: (context, index) {
              return weBuildThings(students[index]);
            },
          ), // ListView.builder
  ); // Scaffold
}
}
```


11:20
DEBUG
Flutter Demo Home Page


11:30
DEBUG
Flutter Demo Home Page
Cassandra Gayle R. Oraiz
Active
BSIT 3
Likes: Wala
1
Ravenclaw
20
Drawing, Gaming, Crochet
Delete
Edit
Harry Potter
Offline
IDK 4
Likes: Heheh
6
Gryffindor
25
Quidditch
Loading Successful!

🏴 Flag 5 — Multiple States

Objective: Make your Student Directory handle multiple UI states correctly.

Your application should now be able to display:

1. Loading
2. Students available
3. No students available

Determine which state should be displayed based on your data.

Test

Verify each state:

Loading
→ Loading UI

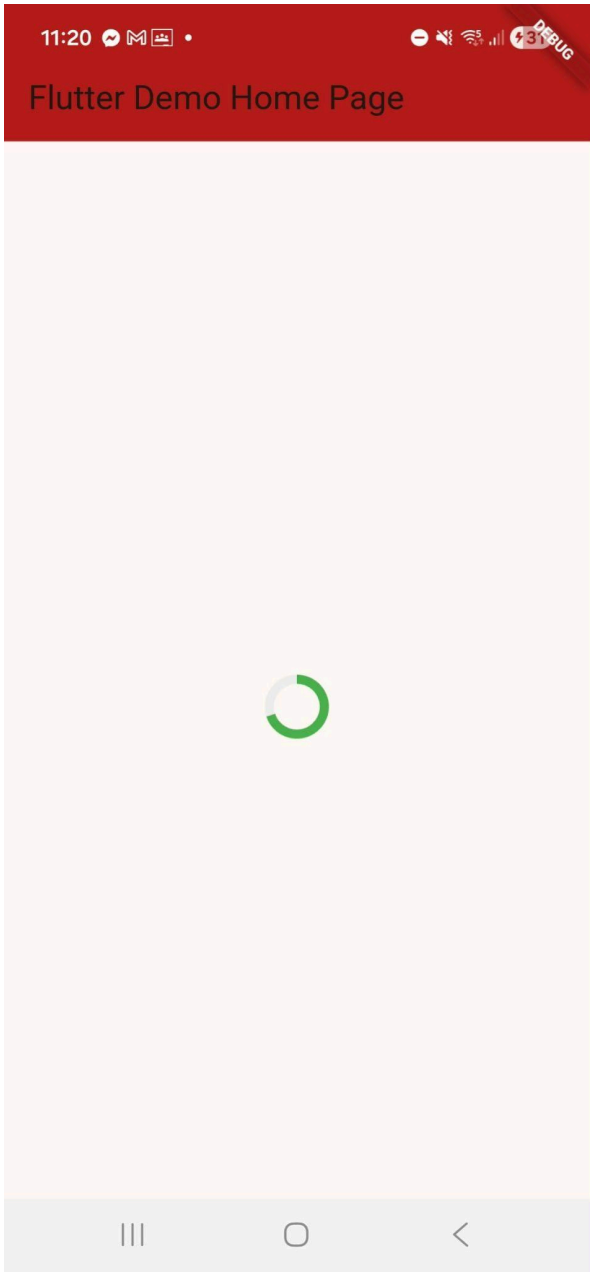
Students available
→ Student List

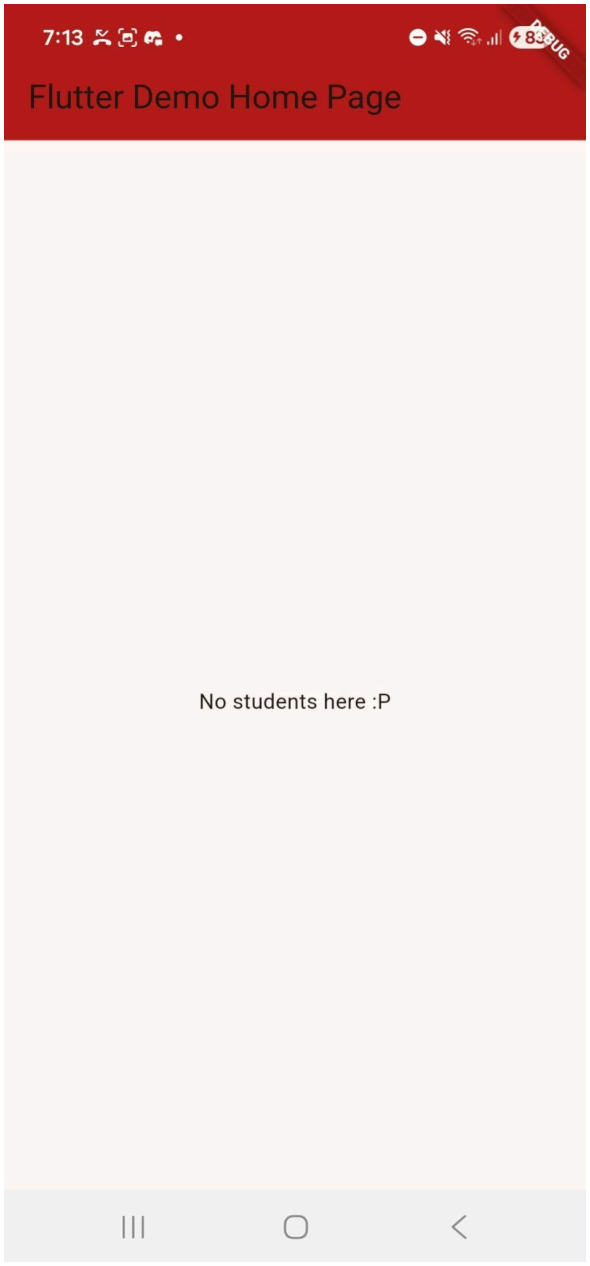
Empty list
→ Empty State

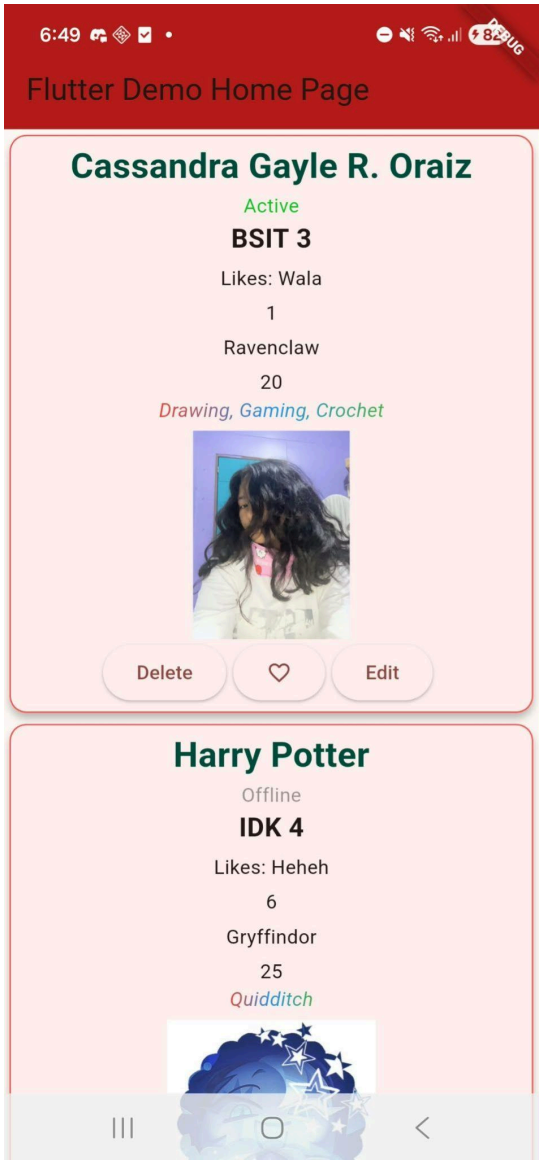
Only the appropriate UI should appear at a time.

Screenshot:

- Loading state
- Student list state
- Empty state


11:20
DEBUG
Flutter Demo Home Page


7:13
DEBUG
Flutter Demo Home Page
No students here :P



Student List State

🏴 Flag 6 — States Work Together

Objective: Make sure conditional UI does not break your existing interactions.

Test the combination of:

- Favorite
- Delete
- Edit
- Active/Inactive
- Show/Hide
- Empty state
- Loading state

Test Scenario

1. Favorite a student.
2. Change or display their Active/Inactive state.
3. Delete students until the directory becomes empty.
4. Verify the Empty State appears.
5. Reload or restore students.
6. Verify the Student Directory appears again.

The application should continue functioning correctly as its state changes.

Screenshot:

- Multiple conditional states working
- Interactive Student Card
- Empty State
- Application functioning normally

3:43

DEBUG

Flutter Demo Home Page

Cassandra Gayle R. Oraiz

Active

BSIT 3

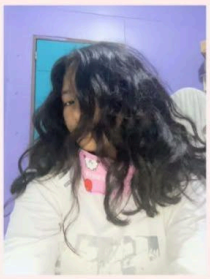
Likes: Wala

1

Ravenclaw

20

Drawing, Gaming, Crochet



Delete ♡ Edit

Pure Vanilla Cookie

Active

BNA 4

Likes: SMC

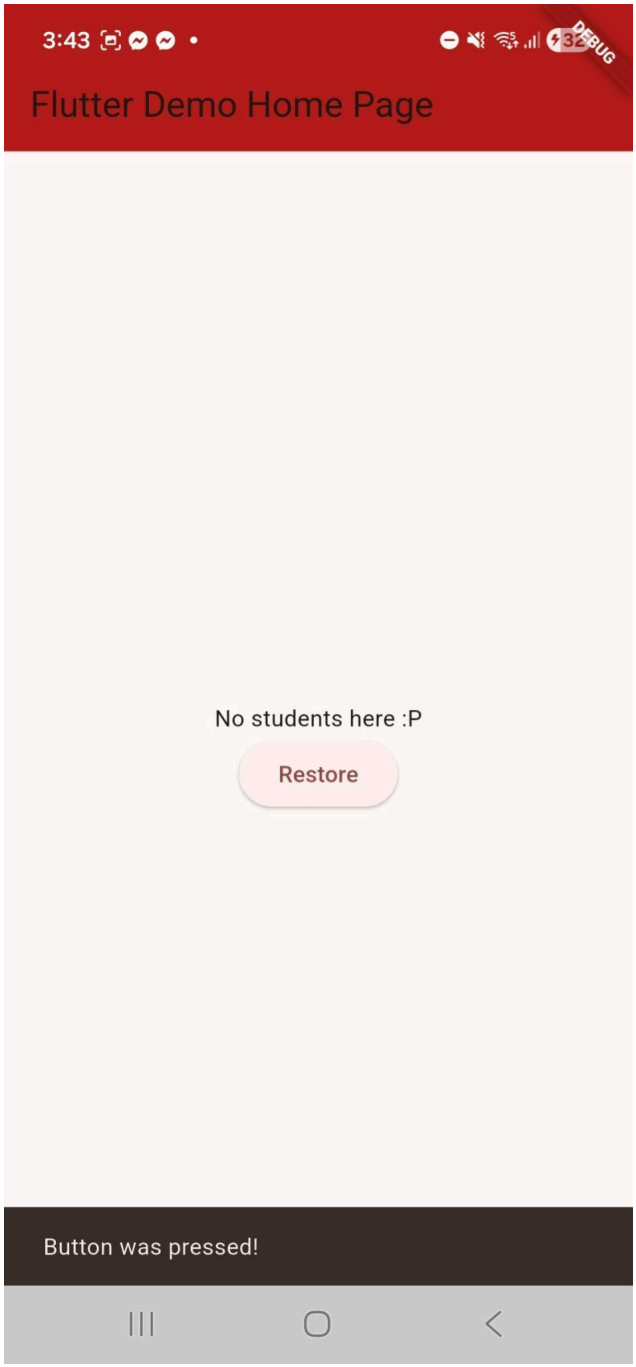
2

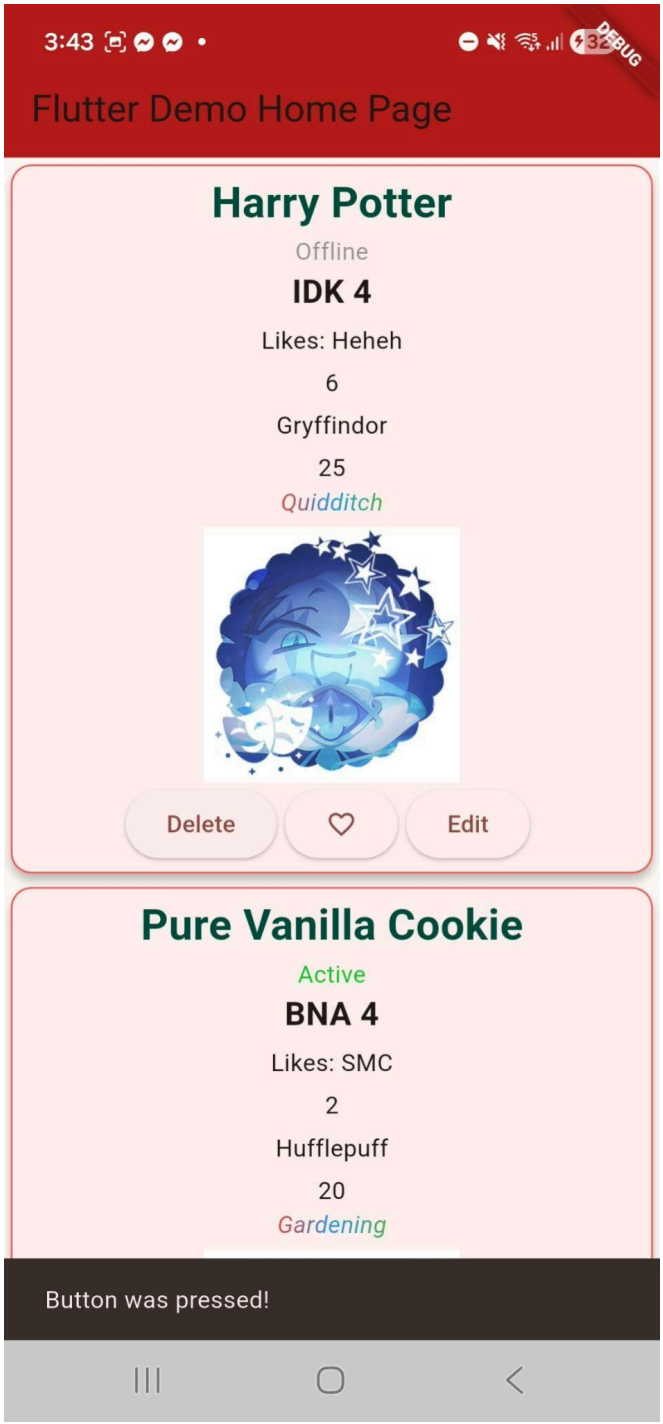
Hufflepuff

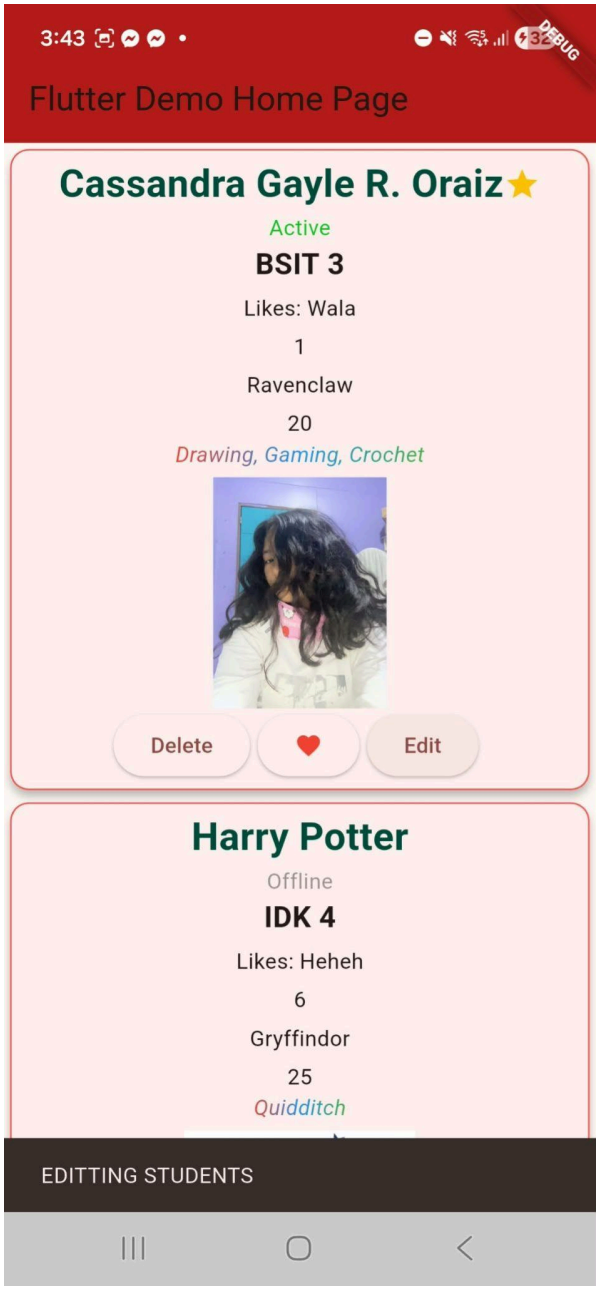
20

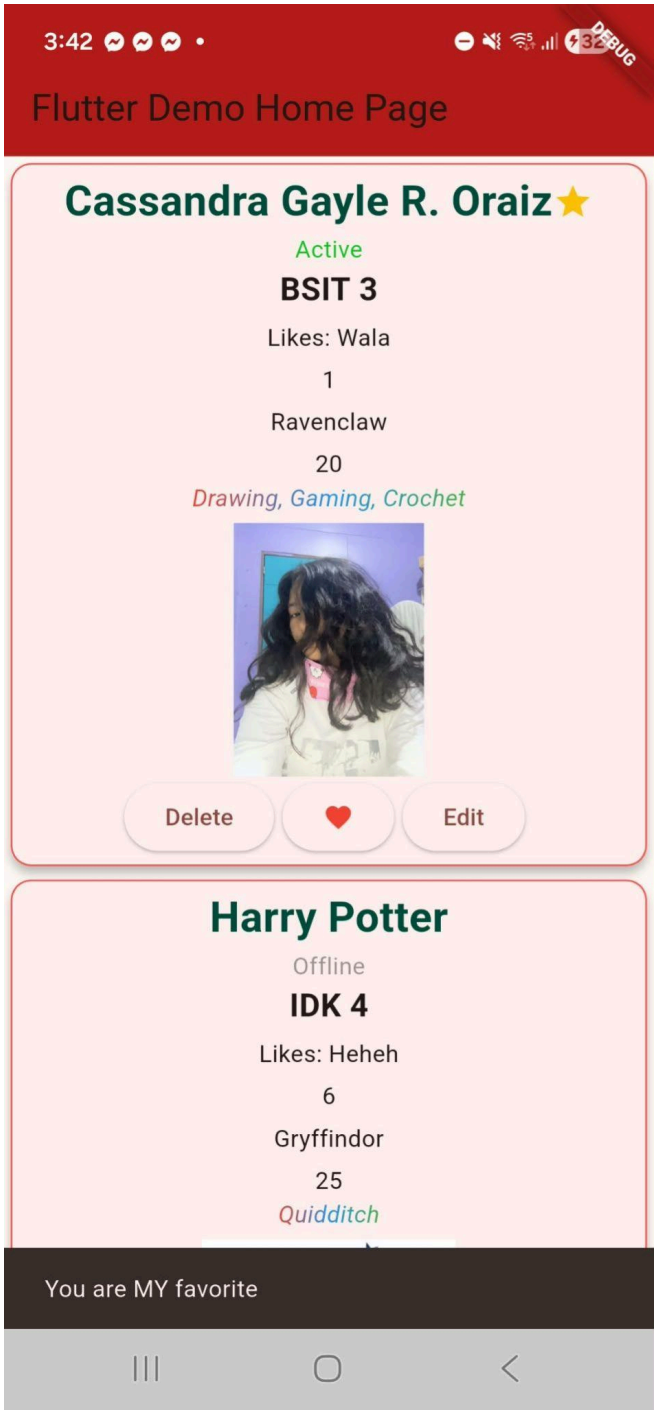
Gardening

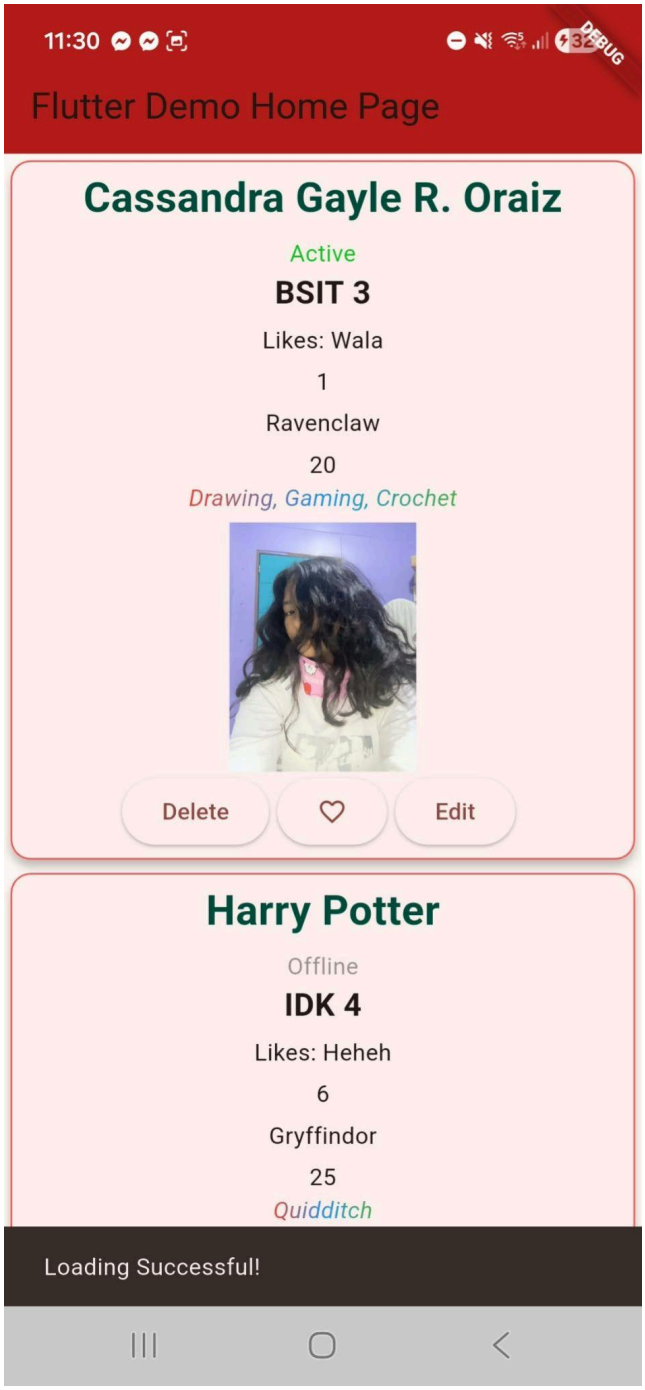
Student Restored


3:43
DEBUG
Flutter Demo Home Page
No students here :P
Restore
Button was pressed!


3:43
DEBUG
Flutter Demo Home Page
Harry Potter
Offline
IDK 4
Likes: Heheh
6
Gryffindor
25
Quidditch
Delete
Edit
Pure Vanilla Cookie
Active
BNA 4
Likes: SMC
2
Hufflepuff
20
Gardening
Button was pressed!


3:43
DEBUG
Flutter Demo Home Page
Cassandra Gayle R. Oraiz
Active
BSIT 3
Likes: Wala
1
Ravenclaw
20
Drawing, Gaming, Crochet
Delete
Edit
Harry Potter
Offline
IDK 4
Likes: Heheh
6
Gryffindor
25
Quidditch
EDITTING STUDENTS


3:42
DEBUG
Flutter Demo Home Page
Cassandra Gayle R. Oraiz
Active
BSIT 3
Likes: Wala
1
Ravenclaw
20
Drawing, Gaming, Crochet
Delete
Edit
Harry Potter
Offline
IDK 4
Likes: Heheh
6
Gryffindor
25
Quidditch
You are MY favorite



Student list loaded. One person is active the other is inactive, star is then hidden.

🏴 Flag 7 — Save Your Progress

Objective: Commit and push your completed Conditional UI Student Directory.

Use a proper Git commit message.

Branch:

`feature/conditional-ui`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

